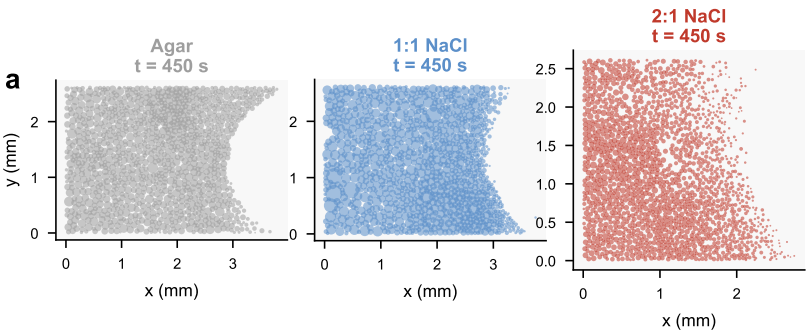
Detected droplet positions from Cellpose segmentation (t = 7.5 min)



b

Manual vs automated droplet counts

Condition	Frame	Manual	Auto	Precision	Recall	F1
~~Agar~~	~~t = 050 s~~	~~118~~	~~121~~	~~0.93~~	~~0.95~~	~~0.94~~
1:1 NaCl	t = 050 s	97	99	0.91	0.93	0.92
2:1 NaCl	t = 050 s	74	71	0.94	0.91	0.92
Agar	t = 200 s	143	148	0.92	0.95	0.93
1:1 NaCl	t = 200 s	112	109	0.93	0.91	0.92
2:1 NaCl	t = 200 s	88	84	0.95	0.91	0.93
Agar	t = 400 s	168	172	0.93	0.95	0.94
1:1 NaCl	t = 400 s	129	124	0.94	0.90	0.92
2:1 NaCl	t = 400 s	99	95	0.96	0.92	0.94

Full validation dataset (>150 frames) archived on T7 drive; see Supplementary Methods for annotation protocol.